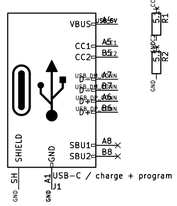
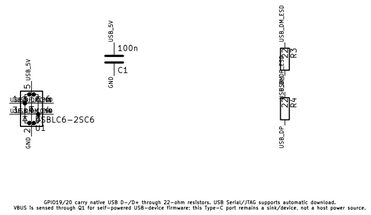
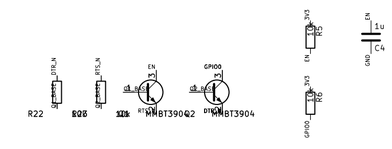
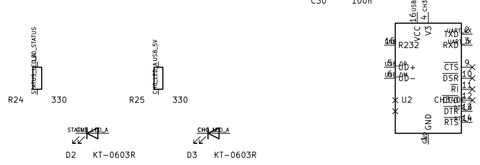
## 1-2 USB-C INPUT, CC SINK, AND ESD

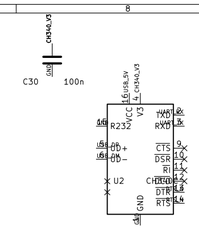



## 3-4 CH340C USB-UART / AUTO PROGRAM

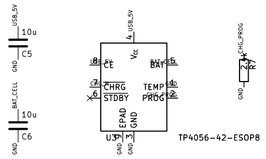


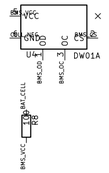



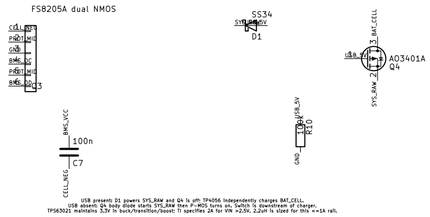



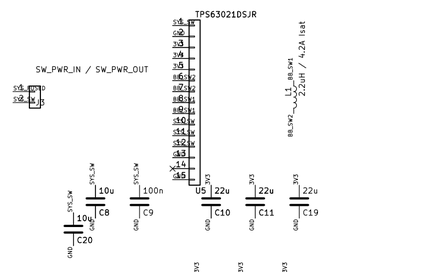

## 5 TP4056 CHARGER (~500 mA)



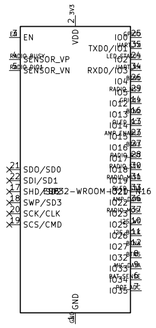

## 6 CELL PROTECTION (DW01A + FS8205A)7-10 LOAD SHARE, PTC, SWITCH, 3.3 V BUCK-BOOST

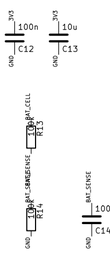


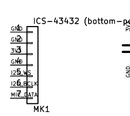



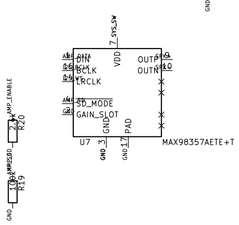



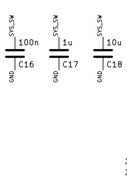

## 11-12 ESP32-WROOM MODULE, ADC, AND SX1262 RADIO 13-15 I2S MICROPHONE, CLASS-D AMP, SPEAKER

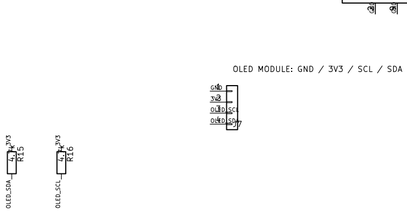


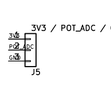



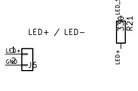


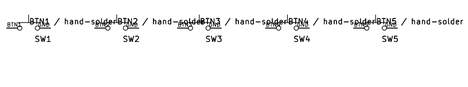
## 16 COMMERCIAL SSD1306 I2C OLED MODULE HEADER

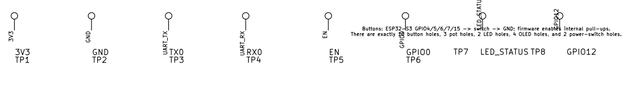
Four through-hole connections only: pin 1 GND, pin 2 3V3, pin 3 SCL, pin 4 SDA.
The commercial module supplies its own SOT23D6 charge-pump parts; no raw SYS_5V reaches OLED logic.

## 17-20 POT, LED, 5× TWO-PIN BUTTONS, TEST PADS






POT_ADC uses ADC1 GPIO2. 100k/100k gives 4.2V -> 2.1V. 2kΩ drain.
and orderly shutdown about 3.0–3.1V loaded. DW01 cutoff is backup only.

See JLC_PART_RESEARCH.md, PIN_ASSIGNMENTS.md, and LAYOUT_NOTES.md.
Power switch must be ON for USB programming; charging works with switch OFF.
TPS63021 fixed buck-boost: 2.2uH; 10uF input; 3x22uF output.
1S Li-ion: WROOM + CH340C USB-UART; SX1262 915MHz; synchronous 3.3V logic
**Custom design**

Sheet: /
File: walkiepcb.kicad_sch

**Title: *ESP32 Walkie-Talkie — Economic PCBA schematic***

| Size: A2 | Date: 2026-08-24 | **Rev: C** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 1/1 |